



Exploring Body Roundness Index's Mediating Role in the Depression-Cognitive Performance Relationship Among US Adults (≥ 60 Years): Evidence From National Health and Nutrition Examination Survey

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Objective Late-life depression is a known risk factor for cognitive decline, yet the role of central adiposity, measured by the body roundness index (BRI), remains unclear. This study examined whether BRI mediates the relationship between depressive symptoms and cognitive performance in older adults.

Methods Our analysis included 2,580 adults aged 60 and over, using data from NHANES. Depressive symptoms were assessed via the Patient Health Questionnaire-9 (PHQ-9), cognitive function was evaluated using the Consortium to Establish a Registry for Alzheimer's Disease (CERAD) Word Learning and Delayed Recall (CERAD-WL/DR), Animal Fluency test (AFT), and Digit Symbol Substitution Test (DSST), and BRI was calculated from waist circumference and height. Analyses included weighted multivariable logistic regression, restricted cubic spline analyses, subgroup and sensitivity analyses, and mediation analyses.

Results Across all assessment modalities, there was a significant link between high PHQ-9 scores and cognitive performance. For continuous PHQ-9 scores, each 1-point increase corresponded to: CERAD (odds ratio [OR]=1.034, 95% confidence interval [CI]: 1.000–1.070, $p=0.049$), AFT (OR=1.046, 95% CI: 1.014–1.079, $p=0.007$), and DSST (OR=1.080, 95% CI: 1.040–1.123, $p<0.001$). Categorical analysis revealed particularly strong effects for moderate-to-severe depression (PHQ-9 ≥ 10) versus minimal symptoms (PHQ-9, 0–4): DSST (OR=2.562, 95% CI: 1.432–4.584, $p=0.004$), AFT (OR=1.834, 95% CI: 1.239–2.714, $p=0.005$), with CERAD showing marginal significance (OR=1.454, 95% CI: 0.936–2.259, $p=0.090$). Sensitivity analyses confirmed the stability of these associations in multiply imputed datasets. BRI exhibited a statistical mediation effect, accounting for 7.23% ($\beta=-0.021$, $p=0.006$) of the depression-DSST association, though mediation effects were nonsignificant for CERAD (6.21%, $p=0.184$) or AFT (4.41%, $p=0.448$).

Conclusion Depression severity was independently linked to lower cognitive performance, with BRI partially mediating this relationship.

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INTRODUCTION

Cognitive decline is a major public health concern affecting the aging population worldwide. In the United States, approximately one in nine adults aged 60 years and older experience cognitive impairment, with numbers projected to rise due to demographic shifts.¹ Among the key psychosocial determinants of late-life cognitive impairment, depression has consistently emerged as a critical risk factor. Numerous longitudinal studies have demonstrated that late-life depression is associated with accelerated cognitive deterioration and increased risk of developing dementia.^{2,3} While the mechanistic pathways linking depression and cognitive decline remain under active investigation, growing attention has been given to intermediary physiological processes, particularly those associated with metabolic dysfunction and body composition. However, few studies have systematically examined whether indices of body fat distribution mediate this relationship in older adults.

The body roundness index (BRI) is an anthropometric indicator designed to capture body shape and visceral adiposity more precisely than traditional measures such as body mass index (BMI) or waist circumference (WC).⁴ BRI's relevance to cognition stems from its ability to capture central adiposity deposition, a source of pro-inflammatory cytokines (e.g., interleukin-6 [IL-6], tumor necrosis factor- α [TNF- α]) that disrupt blood-brain barrier integrity and promote neurodegeneration.⁵⁻⁷ Cross-sectional studies report associations between higher BRI and lower Digit Symbol Substitution Test (DSST) scores.⁸ While numerous studies have documented a positive association between elevated BRI and increased risk of depression,^{9,10} the reverse causal pathway, in which depression contributes to elevated BRI, remains largely unexplored. This represents a critical gap in the literature, given the well-established metabolic consequences of depression. These consequences include weight gain resulting from psychotropic medications, particularly atypical antipsychotics; behavioral mechanisms such as decreased physical activity and sedentary lifestyles; and neuroendocrine alterations, especially leptin resistance, which impair appetite regulation and disrupt energy balance. In addition, no prior study has formally examined whether BRI mediates the relationship between depression and cognitive function using causal inference methodologies. This gap is of particular importance, as depression may worsen metabolic dysfunction through biological mechanisms such as cortisol-induced central adiposity redistribution, thereby initiating a self-reinforcing cycle that accelerates cognitive deterioration.^{11,12} A deeper understanding of these bidirectional interactions is essential for clarifying the complex relationship among mental health, metabolic dysregulation, and cognitive decline in later life.

The National Health and Nutrition Examination Survey (NHANES) offers a unique opportunity to investigate these relationships in a representative sample of U.S. older adults. Drawing on its extensive anthropometric, psychological, and cognitive data, this study seeks to examine whether BRI mediates the relationship between depressive symptoms and lower cognitive performance in adults aged 60 years and older. We hypothesize that higher levels of depression are associated with lower cognitive performance and that BRI statistically mediates this relationship. By integrating body composition metrics into the psychological-cognitive framework, this study contributes to a more nuanced understanding of late-life cognitive aging.

METHODS

Study participants

This study utilized data from the NHANES, a nationally representative program administered by the Centers for Disease Control and Prevention (CDC), which employs a complex, multistage probability sampling design. The initial analytic sample comprised 3,632 adults aged 60 years and older from the 2011–2014 NHANES cycles, which provided comprehensive and validated data on depressive symptoms, cognitive function, and anthropometric indicators, including the BRI. Participants were excluded if data on key covariates, depression status, or cognitive performance were missing. Following the application of these exclusion criteria, the final analysis included 2,580 participants (Figure 1).

All NHANES protocols were approved by the National Center for Health Statistics (NCHS) Research Ethics Review Board, and written informed consent was obtained from all participants prior to data collection. The present study was approved by the Ethics Committee of Taizhou Municipal Hospital (Approval No. LWYJ2025297). As the analysis was conducted using publicly available data, the requirement for informed consent was waived by the ethics committee.

Assessment of cognitive function

The NHANES 2011–2014 modules comprise the Consortium to Establish a Registry for Alzheimer's Disease (CERAD) battery for immediate word list learning and delayed word recall (CERAD Word Learning [CERAD-WL] and CERAD Delayed Recall [CERAD-DR]),¹³ the Animal Fluency test (AFT),¹⁴ and the DSST,¹⁵ which represent key cognitive domains, including memory (both immediate and delayed recall), verbal fluency (semantic memory and word retrieval), and processing speed/executive function (information processing and cognitive flexibility), respectively. These tests were chosen because they are the only standardized cognitive eval-

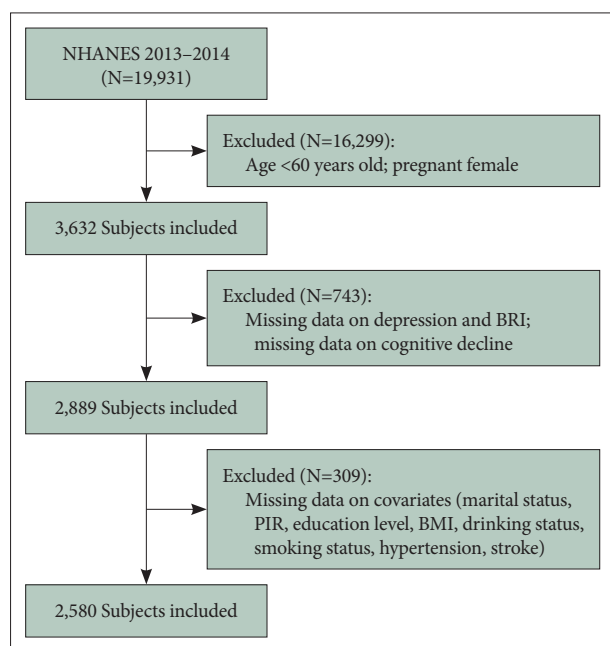


Figure 1. Chart showing the criteria for participant inclusion and exclusion. NHANES, National Health and Nutrition Examination Survey; BRI, body roundness index; BMI, body mass index; PIR, poverty income ratio.

uations available for older adults in the NHANES 2011–2014 dataset.

Focusing on both immediate and delayed recall, the CERAD-WL subtest evaluates verbal learning capacity. Included in the assessment are three consecutive trials for immediate recall and one for delayed recall. Participants are given a list of ten unrelated words during each immediate trial and required to recall as many as they can right after the words are read aloud. Participants perform a delayed recall task around 10 minutes after the final immediate recall trial, recalling the words they were shown earlier without any visual prompts. In CERAD-WL, the total score can be between 0 and 40, with each of the three immediate trials permitting up to 10 points (30 points in total) and the delayed recall trial providing 10 points. For this analysis, the CERAD test was made up of the CERAD-WL and CERAD-DR cognitive performance evaluations.

In the fields of epidemiological and cognitive screening studies, the AFT is commonly used to assess executive functioning, focusing on verbal fluency and cognitive flexibility. Participants are tasked with naming as many different animals as they can within a 1-minute timeframe. Each animal correctly named earns a point, and the total score demonstrates the individual's proficiency in semantic memory retrieval and executive control. In a variety of populations, the AFT has proven to be effective in detecting cognitive decline due to aging.

Through the evaluation of short-term memory, sustained attention, processing speed, and visual scanning, the DSST

offers a detailed assessment of brain health. The DSST was administered in paper format during the examination session. Nine digit-symbol pairs were displayed in a reference key at the top of the test sheet. This key required participants to match each number in a random sequence to its corresponding symbol within a set time limit, assessing processing speed, attention, and executive function. In a span of 2 minutes, participants needed to connect 133 numbers to the appropriate symbols, receiving a point for every accurate match.

Presently, no universally accepted standard exists for the CERAD test (CERAD-WL and CERAD-DR), AFT, and DSST tests to evaluate cognitive decline. Accordingly, we utilized the 25th percentile of the score, the lowest quartile, as the cutoff, consistent with approaches in prior published studies.^{16,17} To assess the robustness of our findings and address potential concerns regarding the dichotomization of data, we will also conduct a sensitivity analysis using continuous test scores, which will allow us to examine the associations without relying on the arbitrary cutoff. By summing the individual scores, a composite score was generated to reflect overall cognitive performance, where higher scores suggest improved cognitive function. Poor cognitive performance was identified by a composite score exceeding 1 standard deviation (SD) below the analytic population's mean,¹⁸ with scores below 47 marking poor performance.

Definitions of depression

The Patient Health Questionnaire-9 (PHQ-9) represents a well-validated, 9-item self-report instrument that operationalizes diagnostic criteria for major depressive disorder as outlined in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition. Made up of nine items, the PHQ-9 is scored from 0 (not at all) to 3 (nearly every day) to measure how often depressive symptoms were experienced in the past 2 weeks. With scores ranging from 0 to 27, higher scores denote more severe symptoms. Participants with a PHQ-9 score of 10 or higher were considered to have moderate to severe depression.¹⁹ Kroenke et al.²⁰ reported that this cutoff has an 88% sensitivity and an 88% specificity for diagnosing major depression. Its concise format and dual functionality for screening and severity evaluation have established it as the most frequently used depression metric in research and clinical settings internationally.

Assessment of BRI

The BRI, an innovative anthropometric measure, calculates body type based on height and WC, assessing central adiposity, where a higher score indicates more fat accumulation. Data on height and WC were obtained from the participants' examination records. Trained health technicians, with the help

of a recorder, measured body dimensions at Mobile Examination Centers to ensure accuracy. Before taking measurements, participants had to remove their clothing and shoes. In an upright position, participants had their standing height measured directly, and their WC was measured at the midpoint between the lower rib and the top of the hips. BRI was computed by $364.2 - 365.5 * (1 - [WC(m)/2\pi]2/[0.5 * height(m)]^2)^{1/4}$.

Covariates

Potential confounding variables adjusted for in the analysis included demographic characteristics (age [continuous, years, mean $\pm$ SD], sex [male/female], race/ethnicity [Mexican American/non-Hispanic Black/non-Hispanic White/other], marital status [married or living with partner/non-married]), socioeconomic factors (education level [less than high school/high school/more than high school], poverty income ratio [PIR: <1/1–3/>3]), lifestyle behaviors (smoking status [never/former/current], alcohol use [never/former/current]), anthropometric measures (BMI [continuous, kg/m², mean $\pm$ SD]), and health conditions (hypertension [yes/no], stroke [yes/no]). Health conditions, including hypertension and stroke, were identified through self-reported physician diagnoses obtained during the household interview. According to NHANES protocol, hypertension is diagnosed when the systolic blood pressure is 130 mm Hg or higher, the diastolic blood pressure is 80 mm Hg or higher, or there is ongoing use of prescribed antihypertensive medication. In a face-to-face interview, stroke was defined by a previous diagnosis reported by the individual and confirmed by a physician. Those who replied “yes” to the question, “Has a medical professional ever told you that you suffered a stroke?” were deemed to have had a stroke. According to U.S. weight classification standards (CDC, 2021), BMI is computed by dividing a person’s weight in kilograms by their height in meters squared.

Statistical analysis

We applied the appropriate NHANES weights to account for the complex survey design, ensuring the results were representative of the U.S. adult population. Specifically, the sampling weights prescribed for the 2011–2014 combined cycles were used, combining the weights from individual survey years to account for the multi-year design. In the analysis, the strata and primary sampling units (PSU) supplied by NHANES were used to adjust for the clustering effects of the survey. The design features were vital for securing unbiased estimates and maintaining the reproducibility of our outcomes. The analyses were performed using logistic regression models that were survey-weighted, adjusting for the complex survey design by including weights, strata, and PSU. For continuous variables, weighted means with SEs were used, and categorical vari-

ables were expressed in terms of weighted proportions. We formulated a series of multivariable survey-weighted logistic regression models for each cognitive domain (CERAD-WL/DR, AFT, and DSST) to examine the relationship between depression severity and cognitive performance. Both continuous variables and categorical indicators representing clinically relevant severity levels were used to analyze PHQ-9 scores. By employing Schoenfeld residuals, we tested the proportional hazards assumption, which was satisfied for all covariates in the Cox models (all $p > 0.05$), ensuring the models were appropriate. To minimize multicollinearity and overadjustment, we conducted a variance inflation factor (VIF) analysis before including covariates in the model, ensuring that only variables with a VIF less than 3 were included, with no variable exceeding this threshold in the final model, thereby minimizing the risk of multicollinearity and overadjustment (Supplementary Table 1). Confounders were identified using prior research and expert knowledge to understand potential influences on the relationship between depressive symptoms and cognitive performance. In order to assess the association between depressive symptoms and cognitive performance, three hierarchical regression models were constructed, with progressively adjusted covariates. The unadjusted crude model, referred to as Model 1, provided an initial estimate of the associations without controlling for potential confounders. To mitigate basic confounding factors, Model 2 was adjusted for demographic variables including age, sex, race, and marital status. To fully account for potential confounders, Model 3 added more lifestyle and health-related factors, such as drinking and smoking status, education level, PIR, BMI, hypertension, and stroke. Odds ratios (ORs) with their respective 95% confidence intervals (CIs) were used to report all estimates. To analyze potential non-linear connections between depressive symptoms and cognitive outcomes, survey-weighted restricted cubic spline (RCS) models with four knots at the 5th, 35th, 65th, and 95th percentiles of the PHQ-9 distribution were developed. To assess the robustness of our findings and mitigate potential biases from the dichotomization of data, we performed a sensitivity analysis using continuous test scores. Subgroup analyses were executed across demographic and clinical layers, and interaction terms were tested to explore effect heterogeneity. For analyzing mediation, logistic regression models were employed to estimate the mediator (BRI) and the outcome (cognitive performance). Using mediation analysis, we assessed the extent to which BRI acted as a mediator in the relationship between depressive symptoms and cognitive performance. Specifically, we evaluated the indirect effect of depression on cognitive performance through BRI, while controlling for relevant covariates such as age, sex, race/ethnicity, and health conditions. By employing bootstrap resampling with 1,000 iterations, we

estimated the CIs for the indirect effects, achieving robust estimates that take into account their non-normal distribution. Chained equations were used for multivariate multiple imputation to deal with missing data, generating 20 imputed datasets with the assumption that the data were missing at random. The robustness of the primary findings was assessed through sensitivity analyses using the imputed datasets. The statistical tests were two-sided, with *p*-values less than 0.05 indicating statistical significance. Analyses were performed with R (version 4.3.2; R Foundation for Statistical Computing).

RESULTS

Baseline characteristics

Table 1 details baseline characteristics based on depressive symptom severity, as measured by the PHQ-9. In the group of 2,580 participants, 75.3% reported minimal symptoms with a PHQ-9 score of 0–4, 15.5% had mild symptoms with scores from 5 to 9, and 9.1% experienced moderate to severe symptoms with scores ranging from 10 to 27. Differences in demographic, socioeconomic, behavioral, and clinical profiles were evident among the various severity groups. The burden of depressive symptoms was greater among younger individuals (mean age: 67.48 vs. 69.09 years; *p*=0.020), those with higher BMI (31.71 vs. 28.68 kg/m²; *p*<0.001), and a higher proportion of females (65.07% vs. 50.93%; *p*<0.001). More severe groups had a higher proportion of Non-Hispanic Black individuals (*p*<0.001). People experiencing more severe symptoms were also more likely to be unmarried, have lower levels of education, and belong to lower-income brackets (all *p*<0.001). The behavior analysis indicated that former drinking increased with symptom severity (*p*<0.001), but smoking status showed no significant difference (*p*=0.88). In clinical terms, an increased prevalence of hypertension and stroke was observed in individuals with moderate to severe depressive symptoms, with *p*-values of 0.040 and less than 0.001, respectively. Cognitive function, evaluated through CERAD, AFT, and DSST, showed a progressive decline with increasing symptom severity (all *p*<0.01).

Associations of depression severity with cognitive performance

Increased severity of depressive symptoms was consistently related to higher chances of lower cognitive performance in every domain assessed. In fully adjusted models (Model 3), each one-point increase in PHQ-9 score was significantly associated with poorer performance on the CERAD test score (OR=1.034, 95% CI: 1.000–1.070, *p*=0.049) (Table 2), AFT score (OR=1.046, 95% CI: 1.014–1.079, *p*=0.007) (Table 3), and DSST score (OR=1.080, 95% CI: 1.040–1.123, *p*<0.001)

(Table 4).

Individuals with moderate to severe depression, as measured by a PHQ-9 score between 10 and 27, had increased odds of experiencing lower cognitive performance compared to those with minimal symptoms, who scored between 0 and 4. The most robust associations were with DSST scores (OR=2.562, 95% CI: 1.432–4.584, *p*=0.004) (Table 4), followed by AFT (OR=1.834, 95% CI: 1.239–2.714, *p*=0.005) (Table 3). Although the connection with CERAD test score in this group wasn't statistically significant (OR=1.454, 95% CI: 0.936–2.259, *p*=0.090) (Table 2), the linear trend across the severity of depression categories was significant (*p* for trend=0.026).

RCS analyses provided additional insights into the dose-response relationships between the severity of depressive symptoms and cognitive performance across multiple domains. The CERAD test score revealed a modest linear association, where increased PHQ-9 scores were linked to higher odds of lower cognitive performance (*p* for overall=0.026), but the non-linear component was not significant (*p* for non-linear=0.303) (Figure 2A). The AFT score showed a similar pattern, where increased depressive symptoms were linearly related to lower cognitive performance (*p* for overall=0.002; *p* for non-linear=0.483) (Figure 2B). DSST scores showed a significant non-linear relationship (overall *p*<0.001; non-linear *p*=0.012), notably indicating that the effect of depressive symptoms on processing speed and executive function may increase past a certain threshold (Figure 2C).

Association between depression severity and cognitive performance among diverse subgroups

Analyses of subgroups showed that the link between the severity of depression and cognitive performance, as indicated by CERAD, AFT, and DSST scores, differed among various demographic and clinical groups. For CERAD, the association was statistically significant among females (OR=1.051, 95% CI: 1.016–1.087, *p*=0.005) and individuals with less than a high school education (OR=1.057, 95% CI: 1.001–1.115, *p*=0.044), with no significant interactions detected for age, sex, race, education, or other covariates (*p* for interaction>0.05) (Supplementary Table 2). In contrast, depression severity was significantly associated with lower AFT scores across nearly all subgroups, including both younger and older adults, males and females, and individuals of diverse racial backgrounds. Notably, the relationship was especially pronounced in participants with less than a high school education (OR=1.069, 95% CI: 1.018–1.122, *p*=0.009), current smokers (OR=1.142, 95% CI: 1.066–1.223, *p*<0.001), and non-drinkers (OR=1.115, 95% CI: 1.046–1.189, *p*=0.002), though none of the interaction terms reached statistical significance (Supplementary Table 3). For DSST scores, a robust and consistent association

with depression severity was observed across all demographic categories, with the effect most prominent in individuals with lower educational attainment (less than high school: OR=1.126,

95% CI: 1.074–1.180, $p<0.001$) and those married or living with a partner (OR=1.120, 95% CI: 1.079–1.162, $p<0.001$) (Supplementary Table 4). Importantly, significant interaction

Table 1. Baseline characteristics of participants stratified by graded severity of depressive symptoms

Characteristic	Total (N=2,580)	PHQ-9 [0–4] (N=1,944)	PHQ-9 [5–9] (N=400)	PHQ-9P [10–27] (N=236)	p
Age (yr)	69.05±0.20	69.09±0.23	69.73±0.41	67.48±0.61	0.020
BMI (kg/m ²)	29.16±0.23	28.68±0.25	30.55±0.57	31.71±0.78	<0.001
Sex					<0.001
Female	1,329 (54.08)	930 (50.93)	251 (65.95)	148 (65.07)	
Male	1,251 (45.92)	1,014 (49.07)	149 (34.05)	88 (34.93)	
Race/ethnicity					<0.001
Mexican American	219 (3.23)	144 (2.70)	40 (4.44)	35 (6.52)	
Non-Hispanic Black	1,264 (80.26)	983 (81.59)	185 (77.21)	96 (71.98)	
Non-Hispanic White	606 (8.11)	445 (7.52)	106 (10.19)	55 (10.42)	
Other	491 (8.40)	372 (8.19)	69 (8.16)	50 (11.09)	
Marital_status					<0.001
Married/living with partner	1,492 (63.52)	1,171 (66.15)	216 (56.77)	105 (48.37)	
Non-married/living with partner	1,088 (36.48)	773 (33.85)	184 (43.23)	131 (51.63)	
Education_level					<0.001
Less than high school	619 (15.22)	420 (13.41)	100 (17.59)	99 (29.97)	
High school	601 (21.58)	436 (20.88)	112 (24.57)	53 (23.36)	
More than high school	1,360 (63.20)	1,088 (65.71)	188 (57.84)	84 (46.68)	
PIR					<0.001
<1	431 (8.91)	273 (7.39)	80 (11.11)	78 (20.86)	
1–3	1,139 (39.02)	835 (36.58)	185 (44.16)	119 (55.13)	
>3	1,010 (52.07)	836 (56.03)	135 (44.73)	39 (24.01)	
Drinking_status					<0.001
Never	396 (12.76)	288 (12.48)	66 (13.86)	42 (13.66)	
Former	731 (23.28)	529 (21.16)	123 (28.78)	79 (35.42)	
Current	1,453 (63.96)	1,127 (66.37)	211 (57.37)	115 (50.92)	
Smoking_status					0.880
Never	1,265 (49.21)	972 (49.72)	188 (48.57)	105 (44.99)	
Former	991 (39.70)	747 (39.33)	153 (40.56)	91 (41.96)	
Current	324 (11.10)	225 (10.95)	59 (10.87)	40 (13.05)	
Hypertension					0.040
No	753 (33.21)	607 (34.95)	93 (27.18)	53 (26.05)	
Yes	1,827 (66.79)	1,337 (65.05)	307 (72.82)	183 (73.95)	
Stroke					<0.001
No	2,395 (93.46)	1,832 (94.76)	362 (90.34)	201 (85.50)	
Yes	185 (6.54)	112 (5.24)	38 (9.66)	35 (14.50)	
CERAD test score	6.27±0.10	6.36±0.10	5.89±0.15	6.05±0.21	0.004
AFT score	18.31±0.20	18.68±0.23	17.35±0.36	16.11±0.44	<0.001
DSST score	52.44±0.57	53.87±0.58	48.12±1.10	45.45±2.14	<0.001

The CERAD test included CERAD-WL and CERAD-DR. Continuous variables were shown in mean±SD and categorical variables were shown in percentages. PHQ-9, Patient Health Questionnaire-9; BMI, body mass index; PIR, poverty income ratio; CERAD, Consortium to Establish a Registry for Alzheimer's Disease; AFT, Animal Fluency test; DSST, Digit Symbol Substitution Test.

effects were noted for race (p for interaction=0.024), education level (p for interaction=0.003), and marital status (p for interaction<0.001), suggesting that the detrimental impact of depressive symptoms on cognitive performance may be moderated by social and educational contexts (Supplementary Table 4).

Mediation analyses-association between depression severity and cognitive performance

Mediation analysis revealed heterogeneous effects of BRI

on the relationship between depressive symptoms and domain-specific cognitive function. For the DSST, BRI statistically mediated the association between depression severity and cognitive performance, with an indirect effect of $\beta=-0.021$ ($p=0.006$), accounting for 7.23% of the total effect (95% CI: 1.14% to 12.46%) (Figure 3). It implies that BRI is partially responsible for the detrimental impact of depression on processing speed and executive function. In contrast, mediation was not statistically significant for the CERAD (indirect effect $\beta=-0.005$, $p=0.184$; proportion mediated: 6.21%) or AFT (indi-

Table 2. Association between depression and lower cognitive performance (CERAD test score) in weighted multivariable logistic regression

Characteristic	Model 1		Model 2		Model 3	
	OR (95% CI)	p	OR (95% CI)	p	OR (95% CI)	p
PHQ-9 score (continuous)	1.028 (0.995–1.062)	0.090	1.048 (1.013–1.083)	0.008	1.034 (1.000–1.070)	0.049
PHQ-9 score (category)						
PHQ-9 [0–4]	Reference		Reference		Reference	
PHQ-9 [5–9]	1.410 (1.078–1.844)	0.014	1.469 (1.035–2.085)	0.032	1.398 (0.985–1.983)	0.059
PHQ-9P [10–27]	1.308 (0.818–2.090)	0.252	1.725 (1.123–2.650)	0.015	1.454 (0.936–2.259)	0.090
p for trend		0.052		0.004		0.026

Model 1 (crude model). Model 2: age, sex, race, marital_status. Model 3: age, sex, race, marital_status, drinking_status, smoking_status, education_level, poverty income ratio, body mass index, hypertension, stroke. CERAD, Consortium to Establish a Registry for Alzheimer's Disease; OR, odds ratio; CI, confidence interval; PHQ-9, Patient Health Questionnaire-9.

Table 3. Association between depression and lower cognitive performance (AFT score) in weighted multivariable logistic regression

Characteristic	Model 1		Model 2		Model 3	
	OR (95% CI)	p	OR (95% CI)	p	OR (95% CI)	p
PHQ-9 score (continuous)	1.060 (1.028–1.093)	<0.001	1.066 (1.030–1.102)	<0.001	1.046 (1.014–1.079)	0.007
PHQ-9 score (category)						
PHQ-9 [0–4]	Reference		Reference		Reference	
PHQ-9 [5–9]	1.542 (1.058–2.246)	0.026	1.457 (0.955–2.224)	0.079	1.366 (0.908–2.053)	0.124
PHQ-9P [10–27]	2.089 (1.414–3.086)	<0.001	2.344 (1.535–3.578)	<0.001	1.834 (1.239–2.714)	0.005
p for trend		<0.001		0.001		0.008

Model 1 (crude model). Model 2: age, sex, race, marital_status. Model 3: age, sex, race, marital_status, drinking_status, smoking_status, education_level, poverty income ratio, body mass index, hypertension, stroke. AFT, Animal Fluency test; OR, odds ratio; CI, confidence interval; PHQ-9, Patient Health Questionnaire-9.

Table 4. Association between depression and lower cognitive performance (DSST score) in weighted multivariable logistic regression

Characteristic	Model 1		Model 2		Model 3	
	OR (95% CI)	p	OR (95% CI)	p	OR (95% CI)	p
PHQ-9 score (continuous)	1.094 (1.059–1.129)	<0.001	1.111 (1.069–1.156)	<0.001	1.080 (1.040–1.123)	<0.001
PHQ-9 score (category)						
PHQ-9 [0–4]	Reference		Reference		Reference	
PHQ-9 [5–9]	2.271 (1.604–3.213)	<0.001	2.274 (1.508–3.430)	<0.001	2.323 (1.529–3.530)	<0.001
PHQ-9P [10–27]	3.100 (1.941–4.952)	<0.001	3.980 (2.161–7.328)	<0.001	2.562 (1.432–4.584)	0.004
p for trend		<0.001		<0.0001		<0.001

Model 1 (crude model). Model 2: age, sex, race, marital_status. Model 3: age, sex, race, marital_status, drinking_status, smoking_status, education_level, poverty income ratio, body mass index, hypertension, stroke. DSST, Digit Symbol Substitution Test; OR, odds ratio; CI, confidence interval; PHQ-9, Patient Health Questionnaire-9.

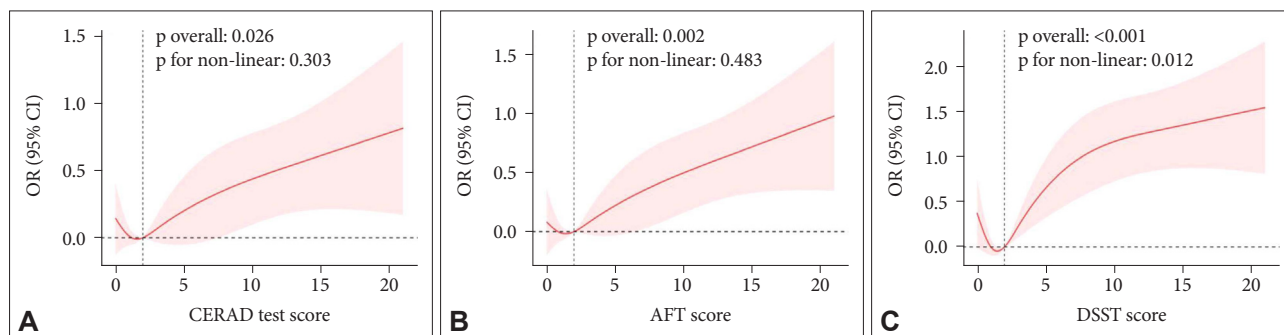


Figure 2. RCS evaluation of the relationship between depression and cognitive decline. Cognitive decline is measured using tests such as CERAD test score (A), AFT score (B), and DSST score (C). OR, odds ratio; CI, confidence interval; CERAD, Consortium to Establish a Registry for Alzheimer's Disease; AFT, Animal Fluency test; DSST, Digit Symbol Substitution Test.

rect effect $\beta = -0.002$, $p = 0.448$; proportion mediated: 4.41%) scores (Figure 3), with wide CIs indicating substantial uncertainty in the mediating role of BRI for memory and verbal fluency outcomes.

Sensitivity analysis

Sensitivity analyses bolstered the reliability of the link between depression severity and cognitive performance. The direction and magnitude of the associations remained consistent across all cognitive domains in multiply imputed datasets. Every one-point rise in the PHQ-9 score was consistently linked to a significant increase in the odds of lower cognitive performance, as measured by CERAD, AFT, and DSST scores (Supplementary Tables 5–7). Upon categorizing depression severity, individuals exhibiting moderate to severe symptoms (PHQ-9 score 10–27) demonstrated a notably higher risk of lower cognitive performance than those with minimal symptoms (score 0–4) (Supplementary Tables 5–7) with the strongest associations observed for DSST, followed by AFT; CERAD results remained directionally consistent though slightly attenuated. To confirm the stability of our results and resolve any concerns about categorizing cognitive performance, we executed a sensitivity analysis using continuous test scores. Model 3, which included adjustments for a full range of covariates, found that depression was still significantly associated with diminished cognitive performance. The depression group had an OR of 2.23 (95% CI: 1.33–3.79, $p < 0.001$) compared to the non-depression group (Supplementary Table 8). The findings of this analysis were in line with the main results, revealing a significant connection between depression and lower cognitive function.

DISCUSSION

In this representative sample of U.S. adults aged 60 years and up, we observed that more severe depressive symptoms were significantly associated with a higher likelihood of poor

cognitive performance across various domains, including memory, executive function, and processing speed. The strongest associations were observed with the DSST score, followed by the AFT score, and these associations remained consistent across subgroups and sensitivity analyses. Significantly, the BRI, a geometry-based marker of central adiposity, emerged as a partial mediator in the statistical analysis, implying that central adiposity could be related to the association between late-life depression and cognitive performance. These findings suggest the complex relationships between mental health, metabolic health, and cognitive decline with age, highlighting the potential value of addressing both psychological and metabolic factors to help preserve cognitive abilities in aging individuals.

Our study contributes to the growing body of evidence linking depressive symptoms to accelerated impairment in cognitive performance in older adults.^{2,3} Cohort studies conducted previously have consistently indicated that depression occurring in late life elevates the risk of dementia and cognitive decline through both psychological and biological mechanisms. However, the majority of past research has been centered on standard anthropometric indices like as BMI and WC, without considering their potential mediating roles. Our investigation adds to the literature and reveals new insights into the potential impact of visceral adiposity on depression-related lower cognitive performance. Studies from the past have identified associations between elevated BRI and poorer cognitive abilities and between higher BRI and an increased risk of depression,^{8,9} but to our knowledge, no study has explicitly tested BRI's mediating role in the depression-cognition pathway. Furthermore, it is evident that obesity heightens neuroinflammation, insulin resistance, and blood-brain barrier dysfunction, all contributing factors to depression and cognitive decline.^{21–24} Our research confirms that a geometry-derived adiposity measure plays a partial role in mediating these effects. Hence, this study not only supports established links but also moves the field forward by merging psychological and metabolic ideas

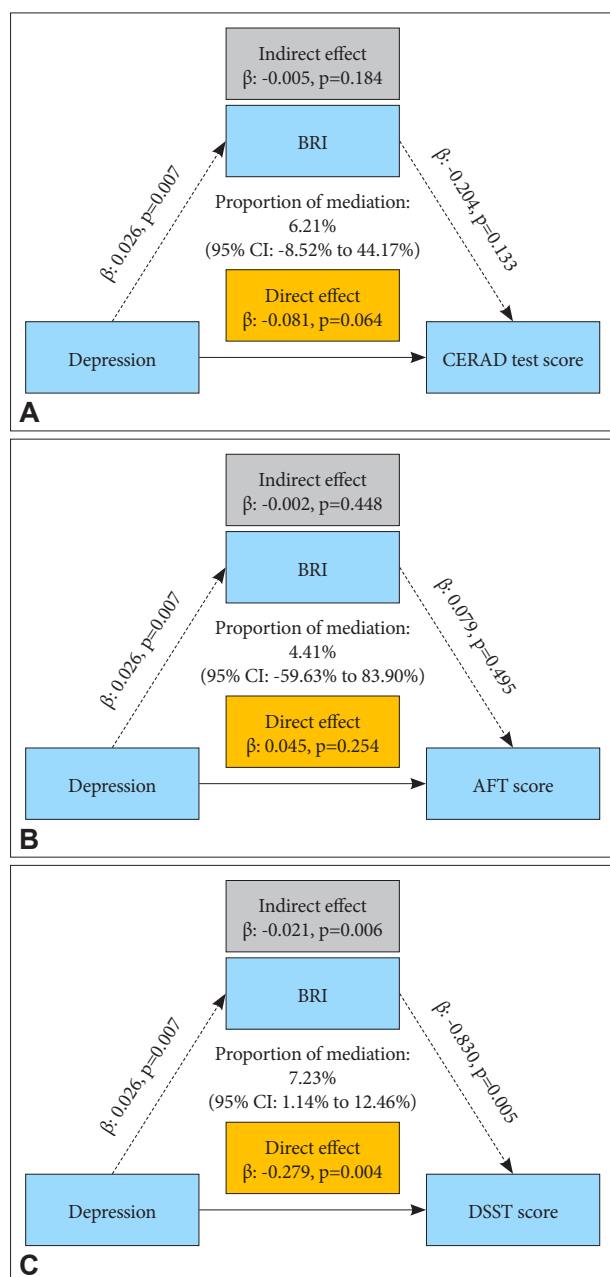


Figure 3. Estimated proportion of the association between depression and cognitive decline mediated by BRI. Cognitive decline is measured using tests such as CERAD test score (A), AFT score (B), and DSST score (C). CI, confidence interval; BRI, body roundness index; CERAD, Consortium to Establish a Registry for Alzheimer's Disease; AFT, Animal Fluency test; DSST, Digit Symbol Substitution Test.

within a causal inference framework.

The BRI could serve as a mediator in the relationship between depression and cognitive decline among older adults, involving several interconnected biological pathways. Central obesity, measured by the BRI, leads to systemic inflammation by increasing the secretion of adipokines like leptin and resistin, along with proinflammatory cytokines such as IL-6 and

TNF- α from visceral adipose tissue.^{7,25-27} The blood-brain barrier can be crossed by these circulating inflammatory mediators, which may negatively impact hippocampal neurogenesis and neural plasticity, thus accelerating cognitive decline in older populations.^{28,29} According to previous research, neuroinflammation resulting from microglial activation and oxidative stress contributes to depressive symptoms and neurodegeneration.²⁹⁻³² Insulin resistance and dyslipidemia, as part of metabolic dysfunction tied to increased BRI, aggravate cerebral glucose hypometabolism and diminish BDNF levels, particularly in memory-related brain regions.³³⁻³⁵ The gut-brain axis may mediate this relationship through BRI-associated alterations in gut microbiota diversity and decreased production of neuroprotective short-chain fatty acids, which have been linked to both depression and cognitive impairment.³⁶⁻³⁸ High-BRI conditions often involve shared endocrine issues, including leptin resistance and a lack of adiponectin, which may directly influence synaptic plasticity and amyloid- β removal and accelerate tau phosphorylation.^{39,40} Moreover, identifying BRI as a potential mediator may support the development of dual-purpose interventions that simultaneously address depressive symptoms and metabolic risk to preserve cognitive health in aging populations.^{12,41-43} According to recent evidence, BRI could be a more sensitive measure of these pathological processes than BMI, due to its enhanced ability to reflect central adiposity distribution.^{44,45} Research in the future should look into whether interventions that target body composition, such as resistance training or adopting a Mediterranean diet, might improve metabolic and cognitive results by breaking this adverse cycle. Our results suggest a meaningful connection between depressive symptoms and cognitive performance in the elderly, with BRI, an indicator of central adiposity, statistically mediating this relationship. We performed a VIF analysis to examine possible collinearity between BMI and BRI, and the results indicated no significant collinearity, with VIF values being far below the threshold of 3. This indicates that including both BMI and BRI in the model did not introduce multicollinearity. In epidemiological studies, BMI is often used to represent overall body fat, while BRI is used to measure central fat, which has a stronger link to metabolic and cognitive health. Given that collinearity is not present, we feel that the inclusion of both variables presents a more thorough view of the connection between body composition and cognitive performance. Although over-adjustment in cross-sectional studies is a concern, our method offers important insights into how body fat affects depression and cognitive function.

This study offers new insights into the mediating role of the BRI in the link between depressive symptoms and lower cognitive performance, but several limitations must be taken into account. The cross-sectional design restricts the ability to draw

causal conclusions, stressing the need for longitudinal studies to clarify the order of events over time. Second, relying on self-reported depression assessments using the PHQ-9 might cause recall or reporting bias, although the instrument's strong psychometric properties lend support to the findings. Third, although a wide variety of covariates were adjusted for, residual confounding could still exist due to unmeasured elements like genetic susceptibility, dietary habits, or inflammation. Fourth, although CERAD and DSST are recognized as valid tools, they might not completely capture every cognitive domain that depression impacts. Fifth, Survey weights were not incorporated into the mediation analysis in this study because of constraints in the software. Despite using survey weights in the primary regression model analyses, the mediation procedure's lack of weight support may constrain the generalizability of the mediation outcomes. Future research could be improved by employing different techniques or software that facilitate weighted mediation analysis, which would better address the intricate sampling design of NHANES.

Conclusion

This study found that more severe depressive symptoms are associated with lower cognitive performance in older U.S. adults, with BRI, a measure of central obesity, statistically mediating this relationship. The findings suggest a role for metabolic dysfunction in the association between depression and cognitive performance, underscoring the need for further research and interventions that consider both psychological and metabolic factors to support cognitive health in aging populations.

Supplementary Materials

The Supplement is available with this article at <https://doi.org/10.30773/pi.2025.0375>.

Availability of Data and Material

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors have no potential conflicts of interest to disclose.

Author Contributions

Conceptualization: Hongwei Liu, Wenzhi Niu, Peng Wei, Zhinan Ye, Caidi He. Data curation: Hongwei Liu, Wenzhi Niu, Peng Wei. Formal analysis: Hongwei Liu, Wenzhi Niu, Peng Wei. Funding acquisition: Zhinan Ye, Caidi He. Investigation: Hongwei Liu, Wenzhi Niu, Peng Wei, Minheng Zhang. Methodology: Hongwei Liu, Wenzhi Niu, Peng Wei, Zhinan Ye. Project administration: Zhinan Ye, Caidi He. Resources: Minheng Zhang, Zhinan Ye, Caidi He. Software: Hongwei Liu, Wenzhi Niu, Peng Wei. Supervision: Zhinan Ye, Caidi He. Validation: Hongwei Liu, Wenzhi Niu, Peng Wei, Zhinan Ye, Caidi He. Visualization: Hongwei Liu, Wenzhi Niu, Peng Wei. Writing—original draft: Hongwei Liu, Wenzhi Niu, Peng Wei. Writing—review & editing: all authors.

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